Exam questions with answers 2024-12-02 ak2030, 36, fak3024, 3137

Vetenskapsteori och vetenskaplig metodik (naturvetenskap) (Kungliga Tekniska Högskolan)



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TaMoS: Previous Exam Questions with Answers

December 2024

This is a document intended for the different KTH courses in the *Theory and Methodology of Science*. It includes questions from each of the three different parts on our exam, along with answer guides. You are not allowed to copy from this text into your own answers; that will be considered as plagiarism.

The document is partitioned into sections that correspond to the respective parts on the exam. After each such section another section follows which includes answer guides to the questions in the former section. Pedagogically speaking, you should try to answer all the questions in a section before looking at the answer guides in the following section. Following this study method enables you to compare your own answers to what is expected of an answer.

Please note that the answer guides vary in detail between the questions. For instance, sometimes when a Part 3 question requires examples from the student's own field of study, the corresponding answer guide will not provide such an example. Instead, you will get more schematic guidelines on what a good answer should contain. Also, note that apart from sometimes being *incomplete* (e.g. by lacking concrete examples) the answers are also sometimes *more detailed* than what is required for getting the highest score or grade, for instance by indicating several different ways of answering some part of the question. Finally, note that here is no "minimum length" for these questions. Altogether, the material provided here should work well as a guide when you try to write and evaluate answers yourself, but it is not to be seen as finished examples of essay answers.

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Part 1 questions (multiple choice)

Question 1. Which of the following claims about observations are true? Mark all correct options.

a) Operationalizations are not needed when phenomena are observed in-directly.

b) Direct observations can be aided.

c) Strictly speaking, if you are using glasses or contact lenses you are doing aided observations.

d) In successful indirect observations you are observing an event causally connected to the feature of interest, rather than the feature of interest itself.

Question 2. In what ways do qualitative and quantitative data differ? Mark all correct options.

a) Large quantitative data sets are easier to summarize than large qualitative data sets.

b) When evaluating qualitative data one should not aim to classify it astrue or false, instead it should be classified as “believable” or “narratively coherent”.

c) Qualitative data is rarely measurable on interval or ratio scales, and therefore there is often no need to specify a unit.

d) When collecting quantitative data, one typically focuses on a few properties, qualitative data on the other hand captures a broad range of properties.

Question 3. What is true about utilitarianism? Mark all correct options.

According to utilitarians...

a) ... it is less important that one acts morally, as compared to having amoral character.

b) ... saving ten people by killing five people could be morally right, everything else being equal.

c) ... an action can be deemed morally right or wrong, regardless of its consequences.

d) ... the morally valuable property is utility, usually characterized as happiness.

Question 4. Which of the following are necessary features of scientific experiments? Mark all correct options.

a) That the hypothesis tested is based on previous research.

b) An effect that can be measured on an interval scale.

c) That the intervention is not done on a representation of a variable, but instead a variable that is actually present.

d) That no auxiliary assumption can be the true reason for a falsifying result, in an experiment.

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Question 5. What is true regarding the virtues of models? Mark all correct options.

a) When comparing two models, valuing precision means choosing the one whose parameter specifications imply the parameter specifications of the other.

b) Having the virtue theoretical tractability means that the model can, in a scientifically meaningful way, be processed by a scientific theory.

c) Transparency means that a model follows from a scientific theory.

d) If the model result is not changed when one of the assumptions are changed, then we can conclude that the model has the virtue simplicity.

Question 6. What are *necessary* conditions for a good operationalization? Mark all correct options.

a) The observable property and the property of interest must be connected in a sufficiently stable way.

b) The observable property and the property of interest must be connected in a valid way, to our best knowledge.

c) Given the practical context, the observation of the observable property must be sufficiently precise.

d) The operationalization must be used to define the property of interest.

Question 7. What is true regarding the steps taken in the methodology of design? Mark all correct options.

a) One step – *validation* – is showing that it fulfils the functional requirements.

b) The methodology of the natural science and the methodology of design consist of the same steps.

c) One step requires that one aims for satisficing by looking for the design that maximizes a certain design function.

d) One step – *validation* – is identifying a possible physical description.

Question 8. One part of the classical definition of knowledge concerns belief. What is true regarding the concept of belief? Mark all correct options.

a) Beliefs can be more or less justified.

b) A person can believe something that is false.

c) There are no degrees when it comes to belief, a person either believes something with certainty or not at all.

d) Beliefs are personal – they are something that a person has.

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Question 9. What is the relationship between scientific *methods* and *methodology*? Mark all correct options.

a) When ordinary methods fail to provide observational data, methodology can do so instead.

b) A number of methods in a given scientific discipline can be said to jointly constitute the methodology of that discipline.

c) Methodology is a way to justify why you do science in a certain way and not in another - methods is how you do it.

d) The basis for methodological reasoning is to determine what a proper argument for the selection of a method would be.

Question 10. What is true of learning from models as analogies? Mark all correct options.

a) By referencing the positive analogies, we can justify the use of negative analogies.

b) By referencing the positive analogies, we can justify the use of neutral analogies.

c) Knowledge of a negative analogy is when we know that something is different between the model and the target.

d) If we do not know if something is true for both model and target, then it is a positive analogy.

Question 11. What is true about falsifiable statements? Mark all correct options. A falsifiable statement…

a) … might be false.

b) … might be true.

c) … can be about something that is not directly observable, but will still be testable.

d) … is either true or false, but not both.

Question 12. Which claims are true of Fisher's significance testing? Mark all correct options.

a) When using Fisher's significance testing, the only type of conclusion that can be supported is the rejection of the hypothesis.

b) By changing the study design, and by changing the partition of possible outcomes, the result of Fisher's significance testing can be influenced.

c) In Fisher's significance testing, the p-value is independent of the tested hypothesis.

d) The test hypothesis and the null hypothesis are always tested by Fisher's significance test.

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Question 13. What is true regarding explanatory virtues (what makes explanations powerful)? Mark all correct options.

a) Being less sensitive to background causes than another explanation means that the explanation is more powerful than the other.

b) Relative to the explanandum, relevant contributing causes should be identified in the explanans.

c) When an explanation is easy to understand, it can be said that the explanandum is precise.

d) Precision in the explanans is measured by the extent to which an explanans references the actual causes of the event to be explained.

Question 14. What is an *ad hoc* hypothesis? Mark all correct options.

a) It is when you change a theory so that it is incompatible with empirical data.

b) A consequence of *ad hoc* hypotheses is an increase in falsifiability.

c) A consequence of *ad hoc* modifications is a decrease in falsifiability.

d) It is a change C of a theory T resulting in an updated version T' such that C attempts to include some observations so that T is saved from being falsified.

Question 15. Which of the following claims about experiments and observational studies are correct? Mark all correct options

a) In experiments, one always successfully controls all relevant variables; in observational studies one does not control all relevant variables.

b) In an observational study, one does not intervene.

c) One can perform an experiment in a natural environment.

d) Inferences drawn from experiments and observational studies can be equally justified.

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Part 1 answers

Question 1. a) False. b) True. c) True. d) True.	*Question 6.* a) True. b) True. c) True. d) False.	*Question 11.* a) True. b) True. c) True. d) True.
Question 2. a) True. b) False. c) True. d) True.	*Question 7.* a) True. b) False. c) False. d) False.	*Question 12.* a) True. b) True. c) False. d) False.
Question 3. a) False. b) True. c) False. d) True.	*Question 8.* a) True. b) True. c) False. d) True.	*Question 13.* a) True. b) True. c) False. d) False.
Question 4. a) False. b) False. c) True. d) False.	*Question 9.* a) False. b) False. c) True. d) True.	*Question 14.* a) False. b) False. c) True. d) True.
Question 5. a) True. b) True. c) False. d) False.	*Question 10.* a) False. b) True. c) True. d) False.	*Question 15.* a) False. b) True. c) True. d) True.

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Part 2 questions

Problem 1 Experiments

Consider the following description of an experimental study. “The 100 plants were first stratified into three separate height categories, which were then divided into control and treatment group by tossing a coin. If the coin showed heads, the plant was put in the control group; if it showed tails, the plant was inspected and those that showed no sign of malnourishment were put in the intervention group. The plants were then placed in a specially designed greenhouse with all background factors carefully controlled. The researcher continuously gave the treatment group the new fertilizer X, while the control group got a standard fertilizer. Once a week, the same researcher inspected the growth of the plants.”

Does this describe a lab experiment or a field experiment? Motivate you answer.

Next, the division into control and treatment group in this experiment is poorly performed. Explain why and suggest how it could be improved.

Finally, discuss how we could implement researcher blinding and whether that would improve the experiment.

Problem 2 Explanations

Construct a causal explanation that is not a DN explanation, using the following question as explanandum: “Why did Eric succeed on the course’s final exam last month?”. Try to find a plausible explanation – however, we will not evaluate your answer for its accuracy.

Next, make some changes to improve the precision of the explanandum and then the precision of the explanans. Explain why they are improvements.

Finally, discuss to what degree the new explanation is cognitively salient for a target audience of 10-year-olds and university students respectively.

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Part 2 answer guides

Problem 1 Experiments

Consider the following description of an experimental study. "The 100 plants were first stratified into three separate height categories, which were then divided into control and treatment group by tossing a coin. If the coin showed heads, the plant was put in the control group; if it showed tails, the plant was inspected and those that showed no sign of malnourishment were put in the intervention group. The plants were then placed in a specially designed greenhouse with all background factors carefully controlled. The researcher continuously gave the treatment group the new fertilizer X, while the control group got a standard fertilizer. Once a week, the same researcher inspected the growth of the plants."

1. *Does this describe a lab experiment or a field experiment? Motivate you answer.* ***(ca 1p)***

2. *Next, the division into control and treatment group in this experiment is poorly performed. Explain why and suggest how it could be improved.* ***(ca 2p)***

3. *Finally, discuss how we could implement researcher blinding and whether that would improve the experiment.* ***(ca 2p)***

1. That "[t]he plants were then placed in a specially designed greenhouse with all background factors carefully controlled" indicates that it is a lab experiment. For a field experiment, we would expect less control of background variables (but note that a field experiment might well take place in a greenhouse as well). An intervention is described (applying fertilizer X), indicating that this is indeed an experiment and not an observational study.

2. The crucial flaw is the selection bias that arises when only the treatment group gets an extra inspection for malnourishment, excluding such plants. This introduces an unnecessary difference between control and treatment group, and one can well suspect that initially well-nourished plants respond better to the intervention, which might render a biased result. Other factors could be mentioned as well, for instance the sample size or other background factors that should be controlled for, but the selection bias must be noted for full score on this part.

3. To implement this kind of blinding, one must make sure that the people who observe and interpret the results don't know which plants belong to which group. A standard way of doing this would be to have a third party label each plant and not reveal which labels are for which group and/or mask the fertilizers in the same way. This way, neither the people administering the fertilizer nor the ones evaluating the results will know which group is which. This might improve the experiment. For instance, it is possible that there is some subjectivity when it comes to measuring growth (note that the question doesn't include any details about *how* it is measured, i.e. operationalized), which might result in confirmation bias; perhaps it involves evaluating how healthy the leaves look. Knowing which plants are in which group might also lead the researchers to give extra care to the treatment group, apart from administering the fertilizer.

   Note that since the details aren't specified in the question, one might also assume that there is an objective measurement process in place with negligible possibilities for biases and interpretation problems to arise. An argument for this can also render full score on this part (but one must still describe how blinding might be performed).

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Problem 2 Explanations

Construct a causal explanation that is not a DN explanation, using the following question as explanandum: "Why did Eric succeed on the course's final exam last month?". Try to find a plausible explanation – however, we will not evaluate your answer for its accuracy. ***(ca 1p)***

Next, make some changes to improve the precision of the explanandum and then the precision of the explanans. Explain why they are improvements. ***(ca 2p)***

Finally, discuss to what degree the new explanation is cognitively salient for a target audience of 10-year-olds and university students respectively. ***(ca 2p)***

Example:
Eric succeeded on the course's final exam last month. (explanandum)
Eric attended all learning activities during the course and before the exam he re-read all the course material, making sure to truly understand it thoroughly.

Starting with the precision of the explanandum, it needs to be improved in the sense of bringing forth a contrast class indicating what (potentially) difference-making causes that a "good" (that is, accurate, precise and non-sensitive) explanans – in other words an answer – should refer to. Only through an explicit contrast class in the explanandum is it possible to understand what causes that are of interest in order to understand the situation. An example could be:

Eric succeeded on the course's final exam, as compared to the same course's mid-course exam.

With this explanandum it's evident that one is interested in the difference between Eric's exam result in the middle of the course and his result at the end of the course. In other words, something must have happened, or been done differently, between these two exams. A successful explanans would correctly point out such a causal difference(s). An example of an improved explanans could therefore be:

Eric attended all learning activities during the course. Before the mid-exam he didn't find time to practice the so-far-covered course material. Before the final exam he spent a week re-reading the course material.

In the last part of this problem, the student is expected to discuss the virtue of cognitively salience and apply it to one's previous, improved, explanation (especially its explanans). This could be done in several ways as cognitively salience is a virtue rather open for interpretation and dependent upon who is intended to receive the explanation. For example, one could argue that an explanation intended for a 10-year old has higher cognitively salience the more it avoids technical language dependent upon concepts that such a person lacks. However, there is also a trade-off between simplifying an explanation in terms of language, and using concepts accurately describing the situation in question. True understanding must somehow grasp reality as it is, and it is not always clear that a simpler version of something achieves this important fact, especially when it comes to causal explanations. Thus, cognitive salience cannot be a virtue that merely increases the simpler (i.e. less information) an explanans has. An answer awarded with full points on this sub-question should indicate understanding of this, or some other aspect of cognitively salience.

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Part 3 questions

Problem 1

Consider inferences drawn from an experiment: if all inferences are deductively valid, does that guarantee internal validity, and (conversely) can the inferences be internally valid without being deductively valid? More broadly speaking, does it matter whether the inferences made in the empirical sciences are deductively valid?

Problem 2

Following the classical definition of knowledge, discuss some examples of (gaining) knowledge from within your field of study, or a field that you are interested in. In particular, motivate how one fulfils the justification criterion; what are the most central methodological steps in order to justify what is considered to be a true theory or proposition in your examples? Are there any field-specific criteria within the field for what a sufficient amount of justification amounts to, and is fallibility a great challenge?

Problem 3 (not for FAK3014)

As discussed in this course, risk assessment through expected value (EV) and expected utility (EU) models fits well with utilitarianism, but what about the other strategies for handling decisions under uncertainty? Select at least two of the other strategies and discuss how they fit with utilitarianism and deontology respectively.

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Part 3 answer guides

Below are the Part 3 questions and guidelines to the different parts that an answer should contain. If one of these parts is missing, this can to some extent be compensated by higher quality on other parts of the answer. If too many parts are missing, the answer is considered incomplete.

Problem 1

Consider inferences drawn from an experiment: if all inferences are deductively valid, does that guarantee internal validity, and (conversely) can the inferences be internally valid without being deductively valid? More broadly speaking, does it matter whether the inferences made in the empirical sciences are deductively valid?

There are multiple ways one could argue in this question, but in order to be awarded the highest grade the answer must – apart from providing arguments utilizing correct course terminology in a relevant way – clearly take a position regarding the questions. The following is one way one could argue (although there are certainly other ways as well).

This answer will argue for the general conclusion that inferences having the format of deduction is irrelevant when its content concerns empirical (physical) facts. The main argument for this comes from theory dependence and the Duhem/Quine thesis.

It is always technically possible to construct an argument/reasoning whose form is deductive. That is because the property of deduction is (under one understanding of "deduction") not about the content of the premises, but about their logical form. Here's an example:

Premise 1: If mice can fly then cows can jump dig underground tunnels.
Premise 2: Mice can fly.
Conclusion: Cows can dig underground tunnels.

The conclusion in this argument is clearly false, but that doesn't change the fact that the utilized inference (modus ponens) is deductive, which means that **had** the premises been true then the conclusion would have been guaranteed to be true.

A further fact about the above example is that the premises' content is empirical. That is, the claims provided in the premises – "mice can fly" and "cows can dig underground tunnels" – are claims which, in order to know their truth value, require observations. People cannot simply know that cows cannot dig underground tunnels from the outset (it could be the case that they could!), one has to make observations and generate explanations in order to find out this inability of cows. Contrast such claims with abstract claims: "a triangle has 3 sides". Such claims doesn't require observations. Instead, they can be known with mere abstract reasoning.

Whether the content of premises in an argument utilizing a deductive inference is empirical or abstract matters because if it is empirical, observations are necessary. We know that scientific observations (at least some aided-direct and all indirect ones) are theory dependent, i.e. the observation itself require further theory in order to be justified and reliable. And we also know that scientific claims (e.g. hypotheses) are never tested in isolation due to the fact that any scientific observation hinges on a set contained of further assumptions about the observation (classically one assumes, among other things, that "no measurement errors are present").

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Now, scientific experiments concern empirical questions. Thus, if one constructs one or several inferences based on content/information provided by an experiment, then these inferences – regardless of them being deductive or inductive in their logical form – will contain empirical content. Hence, the above-mentioned problems (theory dependence and Duhem/Quine) will affect such an inference. Importantly, the **way** the inference is affected is in terms of its reliability – one cannot be absolutely certain in the empirical content, even though the inference in question might be deductive. The main advantage of deductive inferences (being truth-preserving, i.e. reliable) has therefore no bearing when the content is empirical.

Problem 2

Following the classical definition of knowledge, discuss some examples of (gaining) knowledge from within your field of study, or a field that you are interested in. In particular, motivate how one fulfils the justification criterion; what are the most central methodological steps in order to justify what is considered to be a true theory or proposition in your examples? Are there any field-specific criteria within the field for what a sufficient amount of justification amounts to, and is fallibility a great challenge?

For the highest grade the student should present, discuss and apply the classical definition of knowledge, justification criteria and fallibility, to some examples of gaining knowledge in some field of study in engineering. Roughly two examples are sufficient, but it depends on how extensive the discussion of each example is.

So, a good start for the answer is to say something along the following lines.

The classical definition of knowledge, as proposed by Plato, is **true justified belief**, and this means that for someone to know something, three conditions must be met: the person must believe it, the belief must be true, and there must be justification for the belief.

Examples from engineering could for example come from civil engineering. Here an application could be that when engineers design a new bridge:

The engineers **believe** that the new bridge design is safe, the bridge must indeed be safe (it must be **true**) under expected loads, and that this can be **justified** by sufficient testing that produces relevant data and evidence, using prototypes, models, simulations and experiments, and by using tools such as The HD-method and statistical hypothesis testing. Discussing and applying the steps of the HD-method in a more detailed way, and or internal- or external validity could also be included if it's done in a relevant way. Aspects and strategies of risk analysis and risk management, such as the precautionary principle, is not required but could also be a good thing to include if it's done in a relevant way.

Another example from engineering could be electrical engineering. Here an application could be that when engineers design a new circuit system applied to wind power plant type X:

The engineers **believe** that the new circuit design will function as intended, and that the circuit must work correctly for real wind power plants of type X (it must be **true**), and that this can be **justified** by testing a prototype system, under various conditions etc. Here many of the same aspects mentioned in the bridge example above, can be appealed for justification. It is however not required to address all of them. The most important thing to focus on is just that there is some kind of testing that provides the justification, and that there is a critical discussion of that using relevant course terminology.

When it comes to the question if there are any field-specific criteria within some field, for what a sufficient amount of justification should amount to, and if fallibility is a great challenge, many answers are possible, depending on the specific field, or example(s) from engineering that are discussed.

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In the case of the bridge, it could be argued that the justification depends on what's at stake, namely human lives. That is, if the bridge is supposed to be used by humans. So, if relevant tests can show that the bridge design can hold for loads well above the least expected human traffic (applying safety factors) then it could be said to be justified that the bridge is safe (for that purpose and specific target of investigation).

In the case of the new circuit system, it could be argued that the justification depends on a certain threshold value, that must be produced by the system. Maybe some efficiency value. If the test data shows that value, then it could be said to be justified that the system works as intended for real wind power plants of type X.

When it comes to fallibility, the technical term should be correctly explained according to the course literature, and correctly applied on the cases discussed.

It can be argued that fallibility indeed is a challenge for both these engineering examples. Even well-justified (that is, well tested) beliefs can be proven wrong from new evidence or better methods. This is captured by the problem of induction (a discussion and explanation of this problem and application of it to some specific field, or examples, is good if done in a relevant way). This is why continuous testing, and updates of standards are essential in engineering fields.

It can also be argued that fallibility is not that big of a problem in one's field, or example(s) because the testing has been so extensive and that the data (e.g. computational and empirical) is, or is almost completely exhaustive, that incorrect knowledge (i.e. that the hypothesis is false) is very unlikely.

It is important that the critical discussion of fallibility contains relevant discussions, examples, applications, explanations and justifications for one's arguments, and (again) that it uses relevant course terminology.

Problem 3 (not for FAK3014)

As discussed in this course, risk assessment through expected value (EV) and expected utility (EU) models fits well with utilitarianism, but what about the other strategies for handling decisions under uncertainty? Select at least two of the other strategies and discuss how they fit with utilitarianism and deontology respectively.

Below are some sketched arguments regarding various strategies. The student must show understanding for the ethical frameworks, the chosen strategies and how one might argue for or against them. Naturally, many other lines of argument are possible. Some ways to elaborate the arguments (for a higher grade) would be to criticize the ethical frameworks or the decision strategies.

Safety factors. Can be seen as a way to be "ethically safe" from a utilitarianist's perspective. Given that some risk of severe harm can be foreseen, and this is given significantly larger weight than economic considerations, one tries to have enough margins so that this probability will be low, even if uncertain events have a larger effect than foreseen. Similarly from a deontologist's perspective, we can say that one "hedges" so that subjects shouldn't stand the risk of harm when they have the right not to be harmed. But of course, it might be tricky when the monetary cost is high and the expected harm is low: would we get better overall consequences by spending the money elsewhere? Furthermore, what duties do we have to prevent possible harmful outcomes, as compared to the duty of avoiding to inflict harm directly?

Principle of insufficient reason. Since this is a special case of the EV model, the argument isn't too different than that one (see course text). Arguably, this principle doesn't fit well with deontology, since one might end up knowingly choosing actions that put people in risk of harm. On the other hand, an utiltarianist with the expected-value approach might see this principle as the best way to go when the probabilities are unknown: performing our due diligence to try to get the least harmful outcome.

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Minimax. This seems to be an utilitarian approach, since it is arguably about choosing smaller harms over larger (at least if we consider 'harm' equal to 'risk of harm'). The strategy is problematic for the deontologist, but perhaps less so than EV/EU models, since here, one actively avoids causing harm, i.e., one might actively steer away from actions that are likely to harm thousands of people, choosing less harmful ones instead. However, it is still problematic for the deontologist that the strategy can be seen as justifying actions that cause harm, even if that harm is less severe than the harm caused by other actions.

Precautionary principle. This is arguably the most straightforward deontological approach, since it is all about avoiding harm. Perhaps some deontologists would propose this strategy even in decisions under risk or ignorance, seeing as the other strategies all seem to be problematic? Conversely, a utilitarianist could easily point at possible gains from taking various actions even if they might have unknown, harmful outcomes, relating to the argument given in the course about this principle being overly restrictive.

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